

TP – Virtualisation with KVM

ING3 - ICC – Cloud Infrastructures

Academic year 2024–2025



- 1 Using **Virt-manager**, create a VM (Virtual Machine) image called `centOS.qcow`. Boot the VM and install CentOS 7, available at the following address:
 - https://buildlogs.centos.org/rolling/7/isos/x86_64/CentOS-7-x86_64-DVD-1511.isoUse the user-mode network stack. Give the machine the hostname `vm1`. Once the OS has been installed and booted, check the VM's IP address and its visibility in the network. That is, its ability to connect to other machines as well as to the host. Can you say who provides the DHCP and DNS services to this VM?
Note: If you are not getting an IP address in your VM, change in `/etc/sysconfig/network-scripts/ifcfg-ens3` the variable `ONBOOT` to `yes` so that it is enabled on boot. ☐
- 2 Clone the `vm1` VM. Boot the new VM with the hostname `vm2` and check its IP address. Configure `vm2`'s network card as a public bridge so that it can be accessed from outside the host (your laptop). Ask someone to ping your VM from another machine. Can you ping your own host from your own VM? ☐
- 3 Create two new images, `vm1prv` and `vm2prv`, from the `vm1` image. Boot these new VMs and configure a private network between them so that they can only see each other. ☐
- 4 Modify `vm1prv` to add a second network card so that it acts as a router and gives internet access to itself and to `vm2prv`. ☐

Versioning

- 5 Take a snapshot from an existing VM. Modify the VM. Go back to the previous version by using the snapshot. ☐

References

1. KVM libvirt assign static guest IP addresses using DHCP on the virtual machine:
<https://www.cyberciti.biz/faq/linux-kvm-libvirt-dnsmasq-dhcp-static-ip-address-configuration-for-guest-os/>
2. Networking - KVM: <http://www.linux-kvm.org/page/Networking>